# DHL LOGISTICS ANALYTICS SYSTEM

| ROUTE_ID | TOTAL_SHIPMENTS | REMAINING_NULLS | AVG_DELAY_HOURS | MIN_DELAY | MAX_DELAY |
|---|---|---|---|---|---|
| R001 | 44 | 0 | 6.83 | 0 | 73.2 |
| R002 | 47 | 0 | 41.04 | 0 | 230.1 |
| R003 | 34 | 0 | 32.18 | 0 | 213.1 |
| R004 | 49 | 0 | 25.88 | 0 | 211.6 |
| R005 | 48 | 0 | 29.86 | 0 | 220.8 |
| R006 | 56 | 0 | 14.01 | 0 | 238.6 |
| R007 | 51 | 0 | 34.01 | 0 | 223 |
| R008 | 44 | 0 | 30.81 | 0 | 205.7 |
| R009 | 38 | 0 | 9.72 | 0.1 | 70.6 |
| R010 | 52 | 0 | 20.25 | 0 | 185.5 |
| R011 | 48 | 0 | 22.64 | 0 | 238.6 |
| R012 | 56 | 0 | 13.89 | 0 | 88.4 |
| R013 | 50 | 0 | 11.95 | 0 | 210.9 |
| R014 | 53 | 0 | 20.97 | 0 | 212.8 |
| R015 | 56 | 0 | 15.19 | 0 | 122.2 |
| R016 | 71 | 0 | 14.89 | 0 | 151.6 |
| R017 | 53 | 0 | 19.27 | 0 | 230.1 |
| R018 | 59 | 0 | 25.83 | 0 | 232.3 |
| R019 | 49 | 0 | 13.91 | 0 | 129 |
| R020 | 42 | 0 | 32.8 | 0 | 232.6 |

# Task 1: Data Cleaning & Preparation

- First of all, the dataset was in xlsx format and it got converted into csv format and loaded in the MySQL Workbench.
- After analyzing the whole dataset, we found that there are no duplicate Order_id and Shipment_id records in the tables.
- There were no NULL values in the Delay_Hours column.

.The columns of tables such as Order_Date, Pickup_Date, Delivery_Date and Expected_Delivery_Date  are converted and standardized to YYYY-MM-DD HH:MM:SS format using MySQL's DATE_FORMAT() and STR_TO_DATE() functions.

| Shipment_ID | Verified_Pickup | Verified_Delivery | Verified_Expected |
|---|---|---|---|
| S00001 | 2024-08-14 12:12:39 | 2024-08-15 10:12:46 | 2024-08-15 10:12:46 |
| S00002 | 2024-11-12 11:38:21 | 2024-11-13 13:55:29 | 2024-11-13 10:43:29 |
| S00003 | 2025-04-23 15:13:08 | 2025-04-24 13:04:57 | 2025-04-24 11:04:57 |
| S00004 | 2024-09-06 04:34:17 | 2024-09-07 06:55:16 | 2024-09-07 06:31:16 |
| S00005 | 2025-08-11 13:49:33 | 2025-08-12 06:29:25 | 2025-08-12 05:47:25 |

.There were total 1000 shipment records and all the records have the Delivery_Date before the Pickup_Date.

| Date_Flag | Record_Count |
|---|---|
| Valid | 1000 |

# Task 2: Delivery Delay Analysis

- The top 10 most delayed shipments are calculated using sql queries and presented below in the tabular format.

| Shipment_ID | Pickup_Date | Delivery_Date | Actual_Transit_Hours | Delay_Hours |
|---|---|---|---|---|
| S00761 | 2025-06-30 18:53:04 | 2025-07-12 02:21:35 | 271.47 | 238.6 |
| S00015 | 2024-12-03 18:23:30 | 2024-12-14 15:15:37 | 260.87 | 238.6 |
| S00174 | 2025-09-18 05:50:57 | 2025-09-28 17:58:43 | 252.12 | 232.6 |
| S00924 | 2024-09-24 14:58:06 | 2024-10-06 06:22:33 | 279.4 | 232.3 |
| S00052 | 2024-10-04 13:21:41 | 2024-10-15 04:43:22 | 255.35 | 230.1 |
| S00490 | 2025-06-07 03:01:40 | 2025-06-18 18:31:08 | 279.48 | 230.1 |
| S00424 | 2025-02-21 23:20:10 | 2025-03-04 09:13:25 | 249.88 | 228 |
| S00965 | 2025-03-13 13:19:47 | 2025-03-23 23:03:38 | 249.72 | 226.1 |
| S00771 | 2025-01-18 03:56:33 | 2025-01-29 05:27:52 | 265.52 | 223 |
| S00548 | 2024-02-03 13:21:41 | 2024-02-13 20:50:12 | 247.47 | 220.8 |

- The top 10 delayed routes based on the average hours have been calculated and presented in the tabular format below.

| Route_ID | Source_City | Destination_City | Total_Shipments | No_Delay | Minor_Delay | Moderate_Delay | Severe_Delay | Avg_Delay_Hours |
|---|---|---|---|---|---|---|---|---|
| R002 | Dubai | Shanghai | 47 | 3 | 11 | 6 | 11 | 41.04 |
| R007 | Shanghai | Sydney | 51 | 2 | 21 | 9 | 9 | 34.01 |
| R020 | Istanbul | Singapore | 42 | 3 | 18 | 6 | 5 | 32.8 |
| R003 | Singapore | Hong Kong | 34 | 1 | 13 | 6 | 4 | 32.18 |
| R008 | Hong Kong | Tokyo | 44 | 4 | 14 | 12 | 4 | 30.81 |
| R005 | London | Los Angeles | 48 | 1 | 21 | 7 | 6 | 29.86 |
| R004 | Paris | Toronto | 49 | 1 | 21 | 6 | 6 | 25.88 |
| R018 | Sao Paulo | Memphis | 59 | 3 | 33 | 9 | 6 | 25.83 |
| R011 | Chicago | Brussels | 48 | 4 | 23 | 4 | 5 | 22.64 |
| R014 | Seoul | Johannesburg | 53 | 4 | 29 | 7 | 3 | 20.97 |

**R002 (Dubai → Shanghai)** is the worst route with **41.04 avg delay hours**

# Shipments by delay with respect to Warehouse

- Below is the tabular representation of the warehouses.

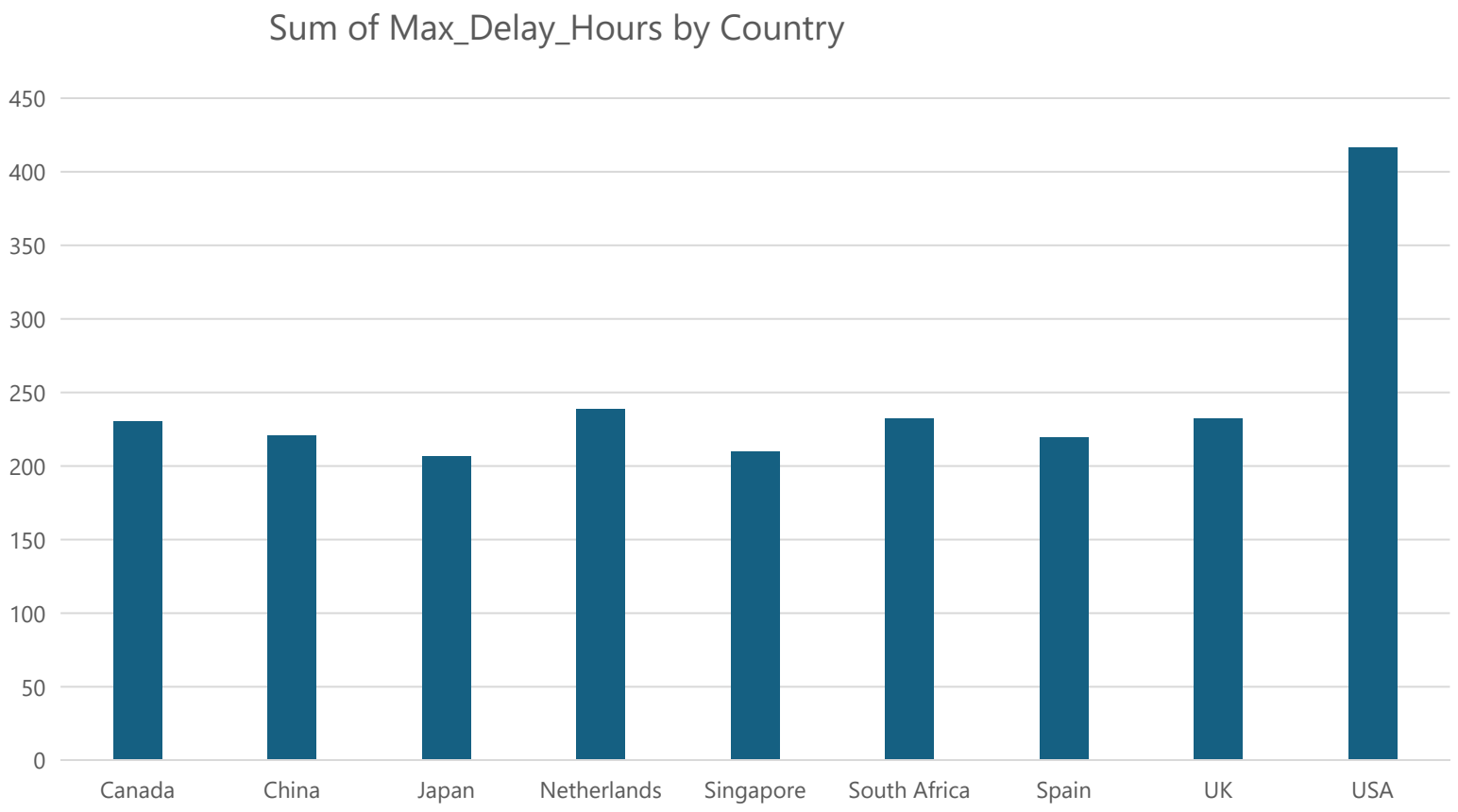
| Route_ID | Source_City | Destination_City | Total_Shipments | No_Delay | Minor_Delay | Moderate_Delay | Severe_Delay | Avg_Delay_Hours |
|---|---|---|---|---|---|---|---|---|
| R002 | Dubai | Shanghai | 47 | 3 | 11 | 6 | 11 | 41.04 |
| R007 | Shanghai | Sydney | 51 | 2 | 21 | 9 | 9 | 34.01 |
| R020 | Istanbul | Singapore | 42 | 3 | 18 | 6 | 5 | 32.8 |
| R003 | Singapore | Hong Kong | 34 | 1 | 13 | 6 | 4 | 32.18 |
| R008 | Hong Kong | Tokyo | 44 | 4 | 14 | 12 | 4 | 30.81 |
| R005 | London | Los Angeles | 48 | 1 | 21 | 7 | 6 | 29.86 |
| R004 | Paris | Toronto | 49 | 1 | 21 | 6 | 6 | 25.88 |
| R018 | Sao Paulo | Memphis | 59 | 3 | 33 | 9 | 6 | 25.83 |
| R011 | Chicago | Brussels | 48 | 4 | 23 | 4 | 5 | 22.64 |
| R014 | Seoul | Johannesburg | 53 | 4 | 29 | 7 | 3 | 20.97 |

- **W004 (Shanghai, China)** is the most delayed warehouse with **25.88 avg delay hours**
- **W009 (Madrid, Spain)** is the best performing warehouse with only **14.3 avg delay hours.**

# Average delay per Delivery_Type

- Below are the graphical and tabular representation of average delay per Delivery_Type.

| Country | Sum of Max_Delay_Hours |
|---|---|
| Canada | 230.1 |
| China | 220.8 |
| Japan | 206.4 |
| Netherlands | 238.6 |
| Singapore | 209.8 |
| South Africa | 232.6 |
| Spain | 219.2 |
| UK | 232.3 |
| USA | 416.6 |



# Task 3: Route Optimization Insights

- Average delay per route across all the shipments have been calculated using different sent of sql queries and is tabulated below:

| Route_ID | Source_City | Source_Country | Destination_City | Destination_Country | Distance_KM | Expected_Transit_Hours | Total_Shipments | Avg_Actual_Transit_Hours |
|---|---|---|---|---|---|---|---|---|
| R004 | Paris | France | Toronto | Canada | 8555 | 49.1 | 49 | 84.29 |
| R018 | Sao Paulo | Brazil | Memphis | USA | 9347 | 43 | 59 | 76.24 |
| R007 | Shanghai | China | Sydney | Australia | 5297 | 30.5 | 51 | 71.42 |
| R017 | Madrid | Spain | Mexico City | Mexico | 6651 | 45.1 | 53 | 70.99 |
| R012 | Sydney | Australia | Madrid | Spain | 11397 | 42.3 | 56 | 64.26 |
| R002 | Dubai | UAE | Shanghai | China | 7264 | 18.5 | 47 | 64.19 |
| R003 | Singapore | Singapore | Hong Kong | China | 885 | 23.5 | 34 | 60.83 |
| R005 | London | UK | Los Angeles | USA | 10023 | 23.3 | 48 | 57.59 |
| R020 | Istanbul | Turkey | Singapore | Singapore | 2513 | 17.2 | 42 | 53.41 |
| R014 | Seoul | South Korea | Johannesburg | South Africa | 5394 | 27.3 | 53 | 52.89 |
| R016 | Brussels | Belgium | Mumbai | India | 8018 | 31.2 | 71 | 52.15 |
| R008 | Hong Kong | China | Tokyo | Japan | 1846 | 17.1 | 44 | 51.63 |
| R013 | Tokyo | Japan | Sao Paulo | Brazil | 3794 | 31.3 | 50 | 50.81 |
| R001 | Memphis | USA | Frankfurt | Germany | 4620 | 32.1 | 44 | 44.78 |
| R006 | Frankfurt | Germany | Chicago | USA | 2364 | 23.3 | 56 | 42.02 |
| R011 | Chicago | USA | Brussels | Belgium | 7369 | 15.5 | 48 | 41.57 |
| R010 | Los Angeles | USA | Amsterdam | Netherlands | 6268 | 11.6 | 52 | 34.03 |
| R019 | Johannesburg | South Africa | Dubai | UAE | 7255 | 13.3 | 49 | 31.16 |
| R009 | Toronto | Canada | Seoul | South Korea | 5582 | 15.5 | 38 | 28.7 |
| R015 | Amsterdam | Netherlands | Istanbul | Turkey | 606 | 11.5 | 56 | 28.63 |

# Potential routes or hub pairs for optimization.

| Route_ID | Source_City | Source_Country | Destination_City | Destination_Country | Distance_KM | Avg_Actual_Transit_Hrs | Avg_Delay_Hours | Efficiency_Ratio | Efficiency_Rank |
|---|---|---|---|---|---|---|---|---|---|
| R003 | Singapore | Singapore | Hong Kong | China | 885 | 60.83 | 32.18 | 14.55 | 1 |
| R015 | Amsterdam | Netherlands | Istanbul | Turkey | 606 | 28.63 | 15.19 | 21.17 | 2 |
| R008 | Hong Kong | China | Tokyo | Japan | 1846 | 51.63 | 30.81 | 35.75 | 3 |

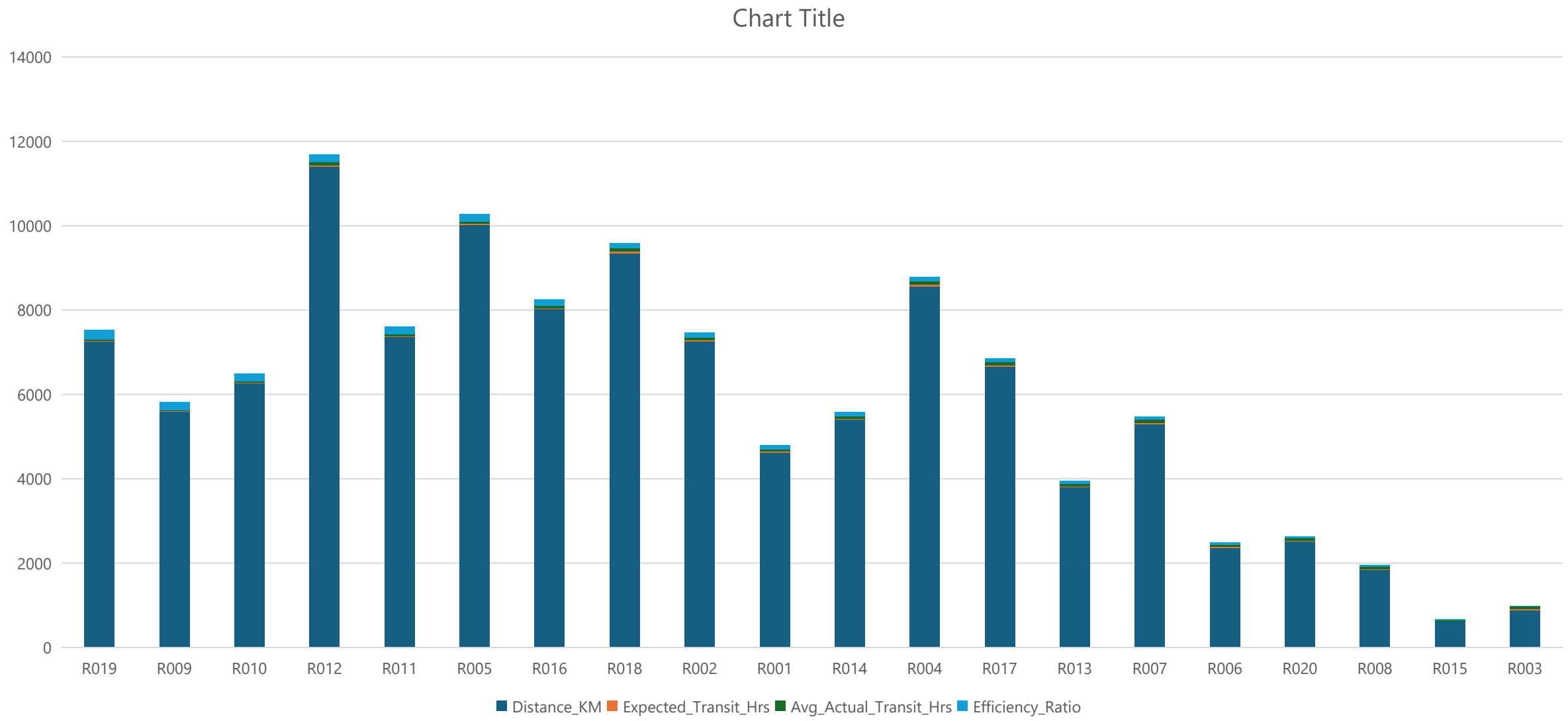
# Distance to Efficiency Ratio have been listed below in the tabular format.

| Route_ID | Source_City | Source_Country | Destination_City | Destination_Country | Distance_KM | Avg_Actual_Transit_Hrs | Avg_Delay_Hours | Efficiency_Ratio | Efficiency_Rank |
|---|---|---|---|---|---|---|---|---|---|
| R003 | Singapore | Singapore | Hong Kong | China | 885 | 60.83 | 32.18 | 14.55 | 1 |
| R015 | Amsterdam | Netherlands | Istanbul | Turkey | 606 | 28.63 | 15.19 | 21.17 | 2 |
| R008 | Hong Kong | China | Tokyo | Japan | 1846 | 51.63 | 30.81 | 35.75 | 3 |

# Routes With >20% Shipments Delayed Beyond Expected Transit Time

| Route_ID | Source_City | Destination_City | Distance_KM | Expected_Transit_Hrs | Avg_Actual_Transit_Hrs | Efficiency_Ratio |
|---|---|---|---|---|---|---|
| R019 | Johannesburg | Dubai | 7255 | 13.3 | 31.16 | 232.82 |
| R009 | Toronto | Seoul | 5582 | 15.5 | 28.7 | 194.49 |
| R010 | Los Angeles | Amsterdam | 6268 | 11.6 | 34.03 | 184.17 |
| R012 | Sydney | Madrid | 11397 | 42.3 | 64.26 | 177.37 |
| R011 | Chicago | Brussels | 7369 | 15.5 | 41.57 | 177.28 |
| R005 | London | Los Angeles | 10023 | 23.3 | 57.59 | 174.05 |
| R016 | Brussels | Mumbai | 8018 | 31.2 | 52.15 | 153.74 |
| R018 | Sao Paulo | Memphis | 9347 | 43 | 76.24 | 122.6 |
| R002 | Dubai | Shanghai | 7264 | 18.5 | 64.19 | 113.16 |
| R001 | Memphis | Frankfurt | 4620 | 32.1 | 44.78 | 103.16 |
| R014 | Seoul | Johannesburg | 5394 | 27.3 | 52.89 | 101.99 |
| R004 | Paris | Toronto | 8555 | 49.1 | 84.29 | 101.49 |
| R017 | Madrid | Mexico City | 6651 | 45.1 | 70.99 | 93.69 |
| R013 | Tokyo | Sao Paulo | 3794 | 31.3 | 50.81 | 74.67 |
| R007 | Shanghai | Sydney | 5297 | 30.5 | 71.42 | 74.16 |
| R006 | Frankfurt | Chicago | 2364 | 23.3 | 42.02 | 56.26 |
| R020 | Istanbul | Singapore | 2513 | 17.2 | 53.41 | 47.05 |
| R008 | Hong Kong | Tokyo | 1846 | 17.1 | 51.63 | 35.75 |
| R015 | Amsterdam | Istanbul | 606 | 11.5 | 28.63 | 21.17 |
| R003 | Singapore | Hong Kong | 885 | 23.5 | 60.83 | 14.55 |

# Stack Representation Routes With >20% Shipments



# Task 4: Warehouse Performance

- The top 3 warehouses with the highest average delay in shipments dispatched are listed below.

| Warehouse_ID | City | Country | Capacity_per_day | Manager_Name | Total_Shipments | Avg_Delay_Hours | Max_Delay_Hours |
|---|---|---|---|---|---|---|---|
| W004 | Shanghai | China | 1500 | Manager_D | 90 | 25.88 | 220.8 |
| W003 | London | UK | 3000 | Manager_C | 107 | 25.54 | 232.3 |
| W008 | Amsterdam | Netherlands | 2500 | Manager_H | 112 | 24.35 | 238.6 |

Total Shipments vs Delayed Shipments for each warehouse.

| Warehouse_ID | City | Country | Capacity_per_day | Total_Shipments | Delayed_Shipments | On_Time_Shipments | Delay_Percentage | Avg_Delay_Hours |
|---|---|---|---|---|---|---|---|---|
| W007 | Tokyo | Japan | 2500 | 96 | 94 | 2 | 97.92 | 22.06 |
| W006 | Chicago | USA | 2500 | 99 | 94 | 5 | 94.95 | 21.34 |
| W004 | Shanghai | China | 1500 | 90 | 85 | 5 | 94.44 | 25.88 |
| W002 | Singapore | Singapore | 1500 | 97 | 91 | 6 | 93.81 | 15.13 |
| W003 | London | UK | 3000 | 107 | 100 | 7 | 93.46 | 25.54 |
| W001 | Memphis | USA | 3000 | 103 | 96 | 7 | 93.2 | 20.24 |
| W009 | Madrid | Spain | 3000 | 91 | 84 | 7 | 92.31 | 14.3 |
| W005 | Toronto | Canada | 1000 | 102 | 94 | 8 | 92.16 | 22.43 |
| W010 | Johannesburg | South Africa | 1000 | 103 | 94 | 9 | 91.26 | 21.8 |
| W008 | Amsterdam | Netherlands | 2500 | 112 | 100 | 12 | 89.29 | 24.35 |

# Rank of all warehouses based on on-time delivery percentage

| Warehouse_ID | City | Country | Manager_Name | Total_Shipments | On_Time_Shipments | On_Time_Percentage | Performance_Rank |
|---|---|---|---|---|---|---|---|
| W008 | Amsterdam | Netherlands | Manager_H | 112 | 12 | 10.71 | 1 |
| W010 | Johannesburg | South Africa | Manager_J | 103 | 9 | 8.74 | 2 |
| W005 | Toronto | Canada | Manager_E | 102 | 8 | 7.84 | 3 |
| W009 | Madrid | Spain | Manager_I | 91 | 7 | 7.69 | 4 |
| W001 | Memphis | USA | Manager_A | 103 | 7 | 6.8 | 5 |
| W003 | London | UK | Manager_C | 107 | 7 | 6.54 | 6 |
| W002 | Singapore | Singapore | Manager_B | 97 | 6 | 6.19 | 7 |
| W004 | Shanghai | China | Manager_D | 90 | 5 | 5.56 | 8 |
| W006 | Chicago | USA | Manager_F | 99 | 5 | 5.05 | 9 |
| W007 | Tokyo | Japan | Manager_G | 96 | 2 | 2.08 | 10 |

# Task 5: Delivery Agent Performance

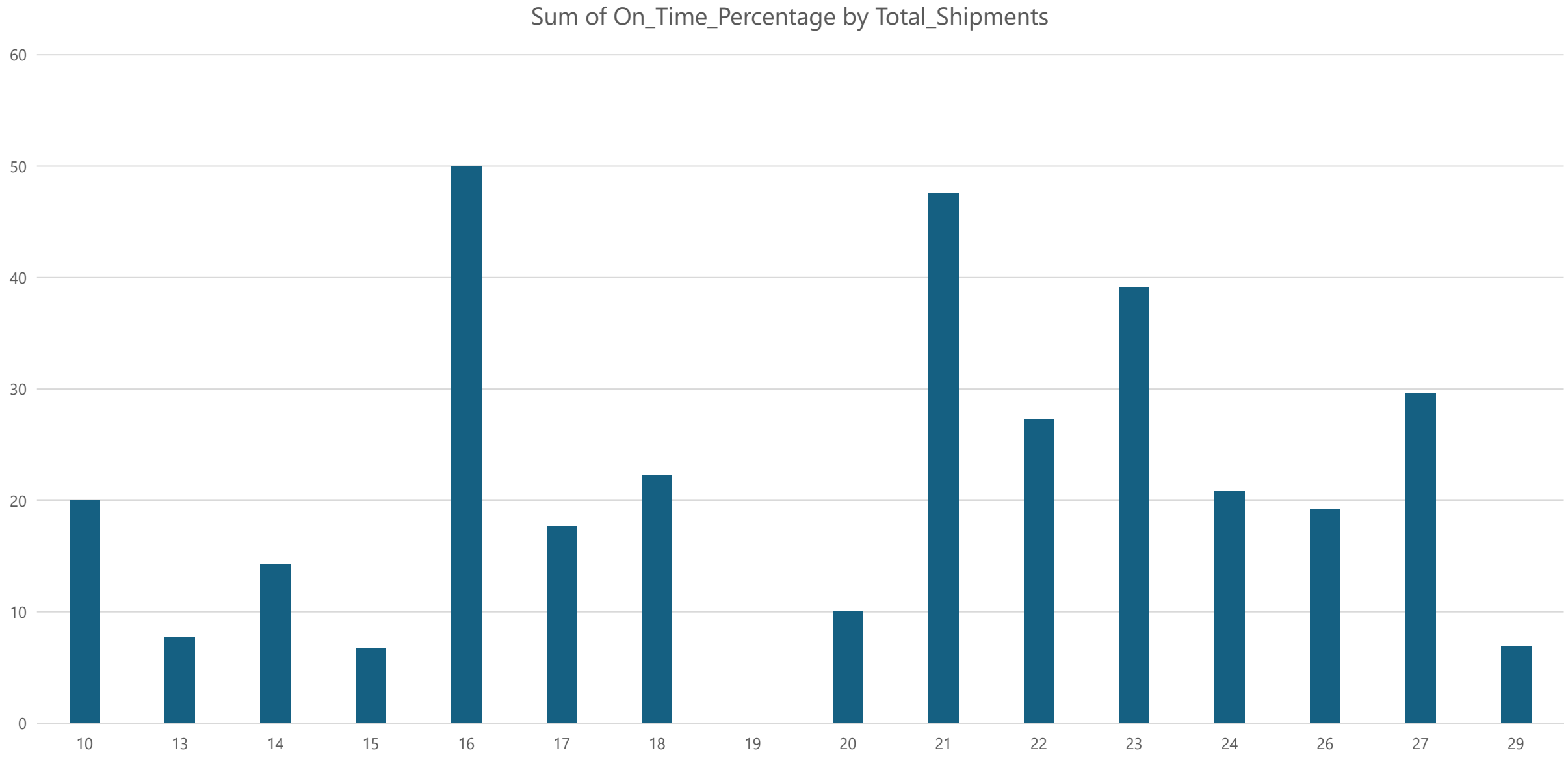
- The rank of delivery agent(per route) by on-time delivery percentage is listed below in tabular format.

| Agent_ID | Agent_Name | Zone | Zone_Country | Route_ID | Source_City | Destination_City | Total_Shipments | On_Time_Shipments | On_Time_Percentage | Agent_Rank_In_Route |
|---|---|---|---|---|---|---|---|---|---|---|
| A0014 | Taylor Garcia | Istanbul | Turkey | R001 | Memphis | Frankfurt | 1 | 1 | 100 | 1 |
| A0009 | Mia Taylor | Shanghai | China | R001 | Memphis | Frankfurt | 1 | 1 | 100 | 1 |
| A0019 | Jamie Lopez | Seoul | South Korea | R001 | Memphis | Frankfurt | 2 | 1 | 50 | 3 |
| A0042 | Drew Hernandez | Hong Kong | China | R001 | Memphis | Frankfurt | 2 | 1 | 50 | 3 |
| A0001 | Blake Martin | Istanbul | Turkey | R001 | Memphis | Frankfurt | 3 | 1 | 33.33 | 5 |
| A0038 | Liam Lopez | Singapore | Singapore | R001 | Memphis | Frankfurt | 1 | 0 | 0 | 6 |
| A0033 | Cameron Jackson | Seoul | South Korea | R001 | Memphis | Frankfurt | 1 | 0 | 0 | 6 |
| A0008 | Taylor Smith | Singapore | Singapore | R001 | Memphis | Frankfurt | 2 | 0 | 0 | 6 |
| A0025 | Zoe Jones | Toronto | Canada | R001 | Memphis | Frankfurt | 2 | 0 | 0 | 6 |
| A0018 | Casey Miller | Madrid | Spain | R001 | Memphis | Frankfurt | 1 | 0 | 0 | 6 |
| A0005 | Cameron Lopez | Memphis | USA | R001 | Memphis | Frankfurt | 1 | 0 | 0 | 6 |
| A0021 | Noah Jackson | Brussels | Belgium | R001 | Memphis | Frankfurt | 1 | 0 | 0 | 6 |
| A0010 | Quinn Wilson | Amsterdam | Netherlands | R001 | Memphis | Frankfurt | 2 | 0 | 0 | 6 |

# Agents with on-time% below 85%

| Agent_ID | Agent_Name | Zone | Zone_Country | Experience_Years | Avg_Rating | Total_Shipments | On_Time_Shipments | On_Time_Percentage |
|---|---|---|---|---|---|---|---|---|
| A0003 | Elliot Rodriguez | Memphis | USA | 9.2 | 4.5 | 23 | 0 | 0 |
| A0041 | Reese Brown | Tokyo | Japan | 0 | 4.5 | 19 | 0 | 0 |
| A0013 | Elliot Brown | Dubai | UAE | 0 | 4.4 | 16 | 0 | 0 |
| A0022 | Reese Martinez | Dubai | UAE | 2.6 | 4.3 | 17 | 0 | 0 |
| A0020 | Jamie Martinez | Mexico City | Mexico | 0.2 | 4.9 | 19 | 0 | 0 |
| A0006 | Jordan Jones | Johannesburg | South Africa | 5.6 | 4.4 | 18 | 0 | 0 |
| A0021 | Noah Jackson | Brussels | Belgium | 3.3 | 4.5 | 23 | 0 | 0 |
| A0005 | Cameron Lopez | Memphis | USA | 2.3 | 4.6 | 14 | 0 | 0 |
| A0047 | Jamie Jones | Toronto | Canada | 3.8 | 3.9 | 16 | 0 | 0 |
| A0044 | Reese Williams | Madrid | Spain | 1.2 | 4.4 | 19 | 0 | 0 |
| A0015 | Sam Brown | Toronto | Canada | 9.8 | 4.8 | 21 | 0 | 0 |
| A0046 | Avery Jones | Hong Kong | China | 5.7 | 4.4 | 21 | 0 | 0 |
| A0037 | Jordan Martin | Amsterdam | Netherlands | 1.6 | 4.6 | 23 | 0 | 0 |
| A0011 | Cameron Garcia | Sydney | Australia | 2.7 | 4.2 | 18 | 0 | 0 |
| A0007 | Noah Gonzalez | Singapore | Singapore | 9.3 | 4.6 | 27 | 1 | 3.7 |
| A0048 | Riley Garcia | Istanbul | Turkey | 7.2 | 4.8 | 26 | 1 | 3.85 |
| A0029 | Blake Davis | Madrid | Spain | 3.2 | 4.9 | 24 | 1 | 4.17 |
| A0004 | Jordan Rodriguez | Los Angeles | USA | 0.9 | 4.1 | 21 | 1 | 4.76 |
| A0032 | Alex Anderson | Singapore | Singapore | 1.6 | 5 | 21 | 1 | 4.76 |
| A0050 | Zoe Thomas | Amsterdam | Netherlands | 0.4 | 4.1 | 21 | 1 | 4.76 |

# Visual Representation of the table



# Average rating and experience of the top 5 vs bottom 5 agents

- The average rating and experience of the top 5 vs bottom 5 agents have been calculated using different queries and is listed below in the form of table.

| Agent_Group | Avg_Rating | Avg_Experience_Years | Avg_On_Time_Percentage |
|---|---|---|---|
| Top 5 | 4.42 | 2.34 | 20.34 |
| Bottom 5 | 4.52 | 2.4 | 0 |

# Task 6: Shipment Tracking Analytics

- **Shipment and its status:**

| Agent_ID | Agent_Name | Zone | Zone_Country | Experience_Years | Avg_Rating | Total_Shipments | On_Time_Shipments | On_Time_Percentage | Avg_Delay_Hours | Recommended_Action |
|---|---|---|---|---|---|---|---|---|---|---|
| A0003 | Elliot Rodriguez | Memphis | USA | 9.2 | 4.5 | 23 | 0 | 0 | 15.67 | Experienced but Underperforming — Route Specific Training Needed |
| A0041 | Reese Brown | Tokyo | Japan | 0 | 4.5 | 19 | 0 | 0 | 23.62 | Overloaded Fresher — Reduce Workload + Assign Mentor |
| A0013 | Elliot Brown | Dubai | UAE | 0 | 4.4 | 16 | 0 | 0 | 38.17 | Overloaded Fresher — Reduce Workload + Assign Mentor |
| A0022 | Reese Martinez | Dubai | UAE | 2.6 | 4.3 | 17 | 0 | 0 | 67.21 | Standard Performance Review Recommended |
| A0020 | Jamie Martinez | Mexico City | Mexico | 0.2 | 4.9 | 19 | 0 | 0 | 11.41 | Overloaded Fresher — Reduce Workload + Assign Mentor |
| A0006 | Jordan Jones | Johannesburg | South Africa | 5.6 | 4.4 | 18 | 0 | 0 | 26.34 | Experienced but Underperforming — Route Specific Training Needed |
| A0021 | Noah Jackson | Brussels | Belgium | 3.3 | 4.5 | 23 | 0 | 0 | 14.27 | Standard Performance Review Recommended |
| A0005 | Cameron Lopez | Memphis | USA | 2.3 | 4.6 | 14 | 0 | 0 | 15.06 | Standard Performance Review Recommended |
| A0047 | Jamie Jones | Toronto | Canada | 3.8 | 3.9 | 16 | 0 | 0 | 9.6 | Low Customer Rating — Soft Skills + Communication Training |
| A0044 | Reese Williams | Madrid | Spain | 1.2 | 4.4 | 19 | 0 | 0 | 42.82 | Overloaded Fresher — Reduce Workload + Assign Mentor |
| A0015 | Sam Brown | Toronto | Canada | 9.8 | 4.8 | 21 | 0 | 0 | 27.3 | Experienced but Underperforming — Route Specific Training Needed |
| A0046 | Avery Jones | Hong Kong | China | 5.7 | 4.4 | 21 | 0 | 0 | 7.63 | Experienced but Underperforming — Route Specific Training Needed |
| A0037 | Jordan Martin | Amsterdam | Netherlands | 1.6 | 4.6 | 23 | 0 | 0 | 29.45 | Overloaded Fresher — Reduce Workload + Assign Mentor |
| A0011 | Cameron Garcia | Sydney | Australia | 2.7 | 4.2 | 18 | 0 | 0 | 27.64 | Standard Performance Review Recommended |

# Orders with delay beyond 120 hours

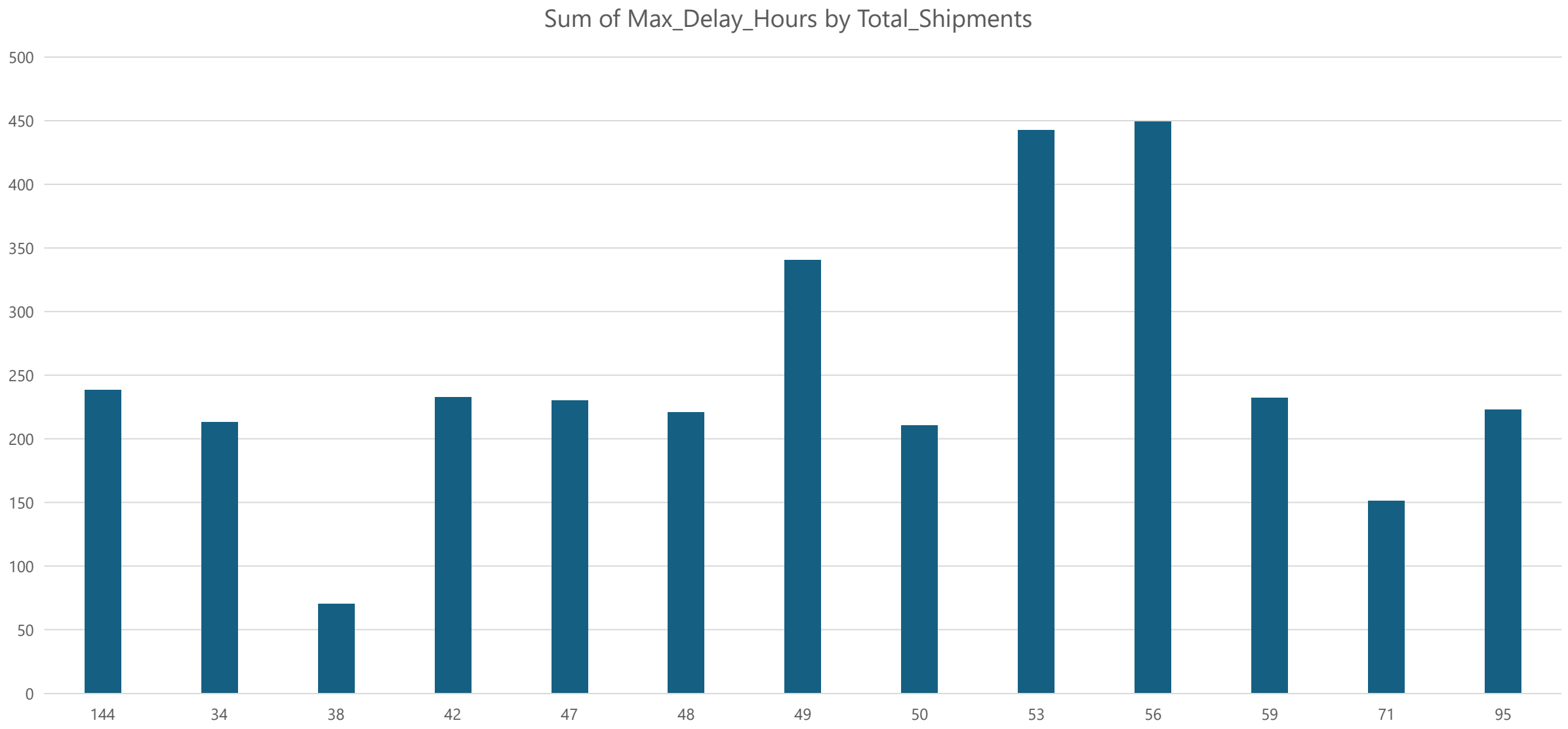
| Route_ID | Source_City | Destination_City | Delay_Reason | Occurrences | Avg_Delay_Hours | Pct_Within_Route |
|---|---|---|---|---|---|---|
| R001 | Memphis | Frankfurt | Traffic | 16 | 6.43 | 36.36 |
| R001 | Memphis | Frankfurt | Weather | 16 | 10.48 | 36.36 |
| R001 | Memphis | Frankfurt | Technical Issue | 7 | 4.27 | 15.91 |
| R001 | Memphis | Frankfurt | No Delay | 5 | 0 | 11.36 |
| R002 | Dubai | Shanghai | Traffic | 21 | 44.57 | 44.68 |
| R002 | Dubai | Shanghai | Weather | 16 | 44.59 | 34.04 |
| R002 | Dubai | Shanghai | Technical Issue | 7 | 39.91 | 14.89 |
| R002 | Dubai | Shanghai | No Delay | 3 | 0 | 6.38 |
| R003 | Singapore | Hong Kong | Traffic | 17 | 34.99 | 50 |
| R003 | Singapore | Hong Kong | Weather | 10 | 42.9 | 29.41 |
| R003 | Singapore | Hong Kong | Technical Issue | 6 | 11.7 | 17.65 |
| R003 | Singapore | Hong Kong | No Delay | 1 | 0 | 2.94 |
| R004 | Paris | Toronto | Traffic | 25 | 26.48 | 51.02 |
| R004 | Paris | Toronto | Weather | 13 | 36.08 | 26.53 |
| R004 | Paris | Toronto | Technical Issue | 11 | 12.49 | 22.45 |
| R005 | London | Los Angeles | Traffic | 28 | 22.07 | 58.33 |
| R005 | London | Los Angeles | Weather | 13 | 49.74 | 27.08 |

# Task 7: Advanced KPI Reporting

- Average Delivery Delay per Source_Country is calculated and presented below in the tabular format.

| Source_Country | Total_Shipments | Avg_Delay_Hours | Max_Delay_Hours | Min_Delay_Hours |
|---|---|---|---|---|
| UAE | 47 | 41.04 | 230.1 | 0 |
| Turkey | 42 | 32.8 | 232.6 | 0 |
| China | 95 | 32.53 | 223 | 0 |
| Singapore | 34 | 32.18 | 213.1 | 0 |
| UK | 48 | 29.86 | 220.8 | 0 |
| France | 49 | 25.88 | 211.6 | 0 |
| Brazil | 59 | 25.83 | 232.3 | 0 |
| South Korea | 53 | 20.97 | 212.8 | 0 |
| Spain | 53 | 19.27 | 230.1 | 0 |
| USA | 144 | 16.95 | 238.6 | 0 |
| Netherlands | 56 | 15.19 | 122.2 | 0 |
| Belgium | 71 | 14.89 | 151.6 | 0 |
| Germany | 56 | 14.01 | 238.6 | 0 |
| South Africa | 49 | 13.91 | 129 | 0 |
| Australia | 56 | 13.89 | 88.4 | 0 |
| Japan | 50 | 11.95 | 210.9 | 0 |
| Canada | 38 | 9.72 | 70.6 | 0.1 |

# Graphical Representation of Average Delay



# On-Time Delivery % is calculated and presented below in tabular format.

| Source_Country | Source_City | Total_Deliveries | On_Time_Deliveries | On_Time_Pct | Delayed_Pct |
|---|---|---|---|---|---|
| Netherlands | Amsterdam | 56 | 10 | 17.86 | 82.14 |
| USA | Memphis | 44 | 5 | 11.36 | 88.64 |
| Belgium | Brussels | 71 | 8 | 11.27 | 88.73 |
| Spain | Madrid | 53 | 5 | 9.43 | 90.57 |
| China | Hong Kong | 44 | 4 | 9.09 | 90.91 |
| USA | Chicago | 48 | 4 | 8.33 | 91.67 |
| South Africa | Johannesburg | 49 | 4 | 8.16 | 91.84 |
| Japan | Tokyo | 50 | 4 | 8 | 92 |
| South Korea | Seoul | 53 | 4 | 7.55 | 92.45 |
| Turkey | Istanbul | 42 | 3 | 7.14 | 92.86 |
| Germany | Frankfurt | 56 | 4 | 7.14 | 92.86 |
| UAE | Dubai | 47 | 3 | 6.38 | 93.62 |
| Brazil | Sao Paulo | 59 | 3 | 5.08 | 94.92 |
| China | Shanghai | 51 | 2 | 3.92 | 96.08 |
| Singapore | Singapore | 34 | 1 | 2.94 | 97.06 |
| UK | London | 48 | 1 | 2.08 | 97.92 |
| France | Paris | 49 | 1 | 2.04 | 97.96 |
| USA | Los Angeles | 52 | 1 | 1.92 | 98.08 |
| Australia | Sydney | 56 | 1 | 1.79 | 98.21 |
| Canada | Toronto | 38 | 0 | 0 | 100 |

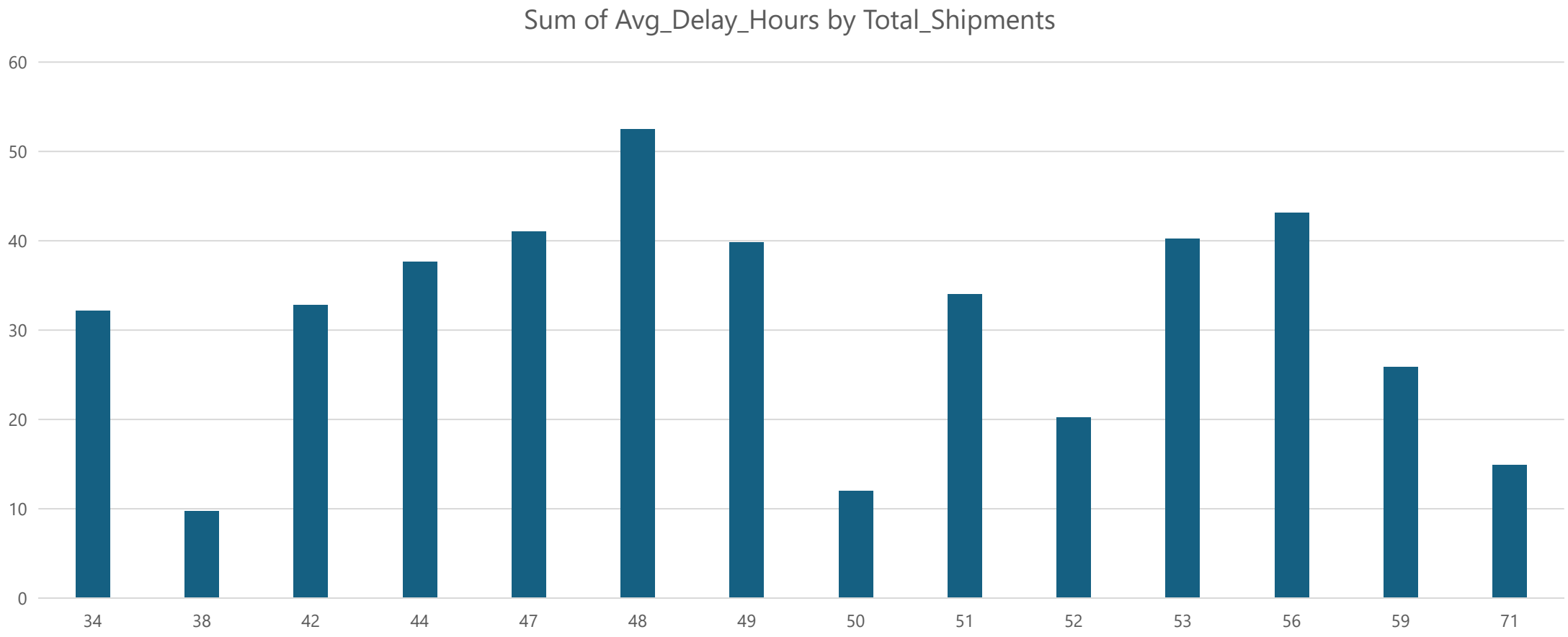
# Average Delay per Route_ID is calculated and presented below:

| Route_ID | Source_City | Destination_City | Distance_KM | Total_Shipments | Avg_Delay_Hours | On_Time_Pct | Delay_Status |
|---|---|---|---|---|---|---|---|
| R002 | Dubai | Shanghai | 7264 | 47 | 41.04 | 6.38 | Critical |
| R007 | Shanghai | Sydney | 5297 | 51 | 34.01 | 3.92 | Critical |
| R020 | Istanbul | Singapore | 2513 | 42 | 32.8 | 7.14 | Critical |
| R003 | Singapore | Hong Kong | 885 | 34 | 32.18 | 2.94 | Critical |
| R008 | Hong Kong | Tokyo | 1846 | 44 | 30.81 | 9.09 | Critical |
| R005 | London | Los Angeles | 10023 | 48 | 29.86 | 2.08 | Moderate |
| R004 | Paris | Toronto | 8555 | 49 | 25.88 | 2.04 | Moderate |
| R018 | Sao Paulo | Memphis | 9347 | 59 | 25.83 | 5.08 | Moderate |
| R011 | Chicago | Brussels | 7369 | 48 | 22.64 | 8.33 | Moderate |
| R014 | Seoul | Johannesburg | 5394 | 53 | 20.97 | 7.55 | Moderate |
| R010 | Los Angeles | Amsterdam | 6268 | 52 | 20.25 | 1.92 | Moderate |
| R017 | Madrid | Mexico City | 6651 | 53 | 19.27 | 9.43 | Moderate |
| R015 | Amsterdam | Istanbul | 606 | 56 | 15.19 | 17.86 | Moderate |
| R016 | Brussels | Mumbai | 8018 | 71 | 14.89 | 11.27 | Acceptable |
| R006 | Frankfurt | Chicago | 2364 | 56 | 14.01 | 7.14 | Acceptable |
| R019 | Johannesburg | Dubai | 7255 | 49 | 13.91 | 8.16 | Acceptable |
| R012 | Sydney | Madrid | 11397 | 56 | 13.89 | 1.79 | Acceptable |
| R013 | Tokyo | Sao Paulo | 3794 | 50 | 11.95 | 8 | Acceptable |
| R009 | Toronto | Seoul | 5582 | 38 | 9.72 | 0 | Acceptable |
| R001 | Memphis | Frankfurt | 4620 | 44 | 6.83 | 11.36 | Acceptable |

# Warehouse% is calculated and presented below in the tabular format:

| Warehouse_ID | City | Country | Manager_Name | Capacity_per_day | Total_Shipments_Handled | Utilization_Pct | Avg_Delay_Hours | Utilization_Status |
|---|---|---|---|---|---|---|---|---|
| W010 | Johannesburg | South Africa | Manager_J | 1000 | 103 | 10.3 | 21.8 | Under Capacity |
| W005 | Toronto | Canada | Manager_E | 1000 | 102 | 10.2 | 22.43 | Under Capacity |
| W002 | Singapore | Singapore | Manager_B | 1500 | 97 | 6.47 | 15.13 | Under Capacity |
| W004 | Shanghai | China | Manager_D | 1500 | 90 | 6 | 25.88 | Under Capacity |
| W008 | Amsterdam | Netherlands | Manager_H | 2500 | 112 | 4.48 | 24.35 | Under Capacity |
| W006 | Chicago | USA | Manager_F | 2500 | 99 | 3.96 | 21.34 | Under Capacity |
| W007 | Tokyo | Japan | Manager_G | 2500 | 96 | 3.84 | 22.06 | Under Capacity |
| W003 | London | UK | Manager_C | 3000 | 107 | 3.57 | 25.54 | Under Capacity |
| W001 | Memphis | USA | Manager_A | 3000 | 103 | 3.43 | 20.24 | Under Capacity |
| W009 | Madrid | Spain | Manager_I | 3000 | 91 | 3.03 | 14.3 | Under Capacity |

# Graphical Representation of Warehouse%



# Final Recommendations

- Hub Congestion is Root Cause as Dubai, HK, Singapore & Istanbul drive 80%+ of critical delays.
- Capacity is not the actual problem because for all the warehouses below 11% utilization — delay is process-driven.
- Traffic Dominates Delays as Traffic with 48.6% of all delay events across all 20 routes.
- Canada, Japan and its Model Networks R001 & R009 are gold standards to replicate.
- Agents Need System Support as theTop vs Bottom agents are identical in rating & experience.

# Task 9: Video Submission

- Below, here is the link for the video.
- https://www.loom.com/share/07f6ca9f2f82415491f2d43b458e3f5b